Vasicek vs CR+

Jan Alsterlind

May 2026

1 Introduction

The Vasicek model and CreditRisk+ (CR+) are two established models for credit portfolio risk. Despite significant differences in mathematical structure, the models in practice describe the same systematic credit risk in large homogeneous portfolios. The difference lies mainly in how this risk is represented:

- The Vasicek model uses a latent normally distributed systematic factor with asset correlation ρ.
- CreditRisk+ uses a stochastic default intensity Λ which is assumed to be gamma distributed.

Both the Vasicek model and CreditRisk+ belong to the so-called ASRF class (Asymptotic Single Risk Factor) of portfolio models. This type of model is formulated such that the model contains a limiting case, the large portfolio limit, where idiosyncratic risk is diversified away and credit losses are entirely driven by a common systematic risk factor.1 In the large portfolio limit, the models can be calibrated such that they produce almost identical loss distributions. This can be done either by matching the variance of the common risk driver (the standard method), or by matching a certain quantile in the loss distribution, for example 99 are discussed below.

2 Common Risk Structure

2.1 Vasicek

In the Vasicek model, each borrower defaults if a latent variable falls below a threshold:

$$X = \sqrt{\rho}\, Z + \sqrt{1-\rho}\, \varepsilon,$$

where $Z \sim \mathcal{N}(0,1)$ and $\varepsilon \sim \mathcal{N}(0,1)$.

Default occurs if

$$X < \Phi^{-1}(p),$$

which gives the conditional probability of default

$$p(Z) = \Phi\left(\frac{\Phi^{-1}(p) - \sqrt{\rho}\, Z}{\sqrt{1-\rho}}\right).$$

2.2 CreditRisk+

In CreditRisk+, the number of defaults is modeled as Poisson distributed given an intensity:

$$N \mid \Lambda \sim \text{Poisson}(\Lambda).$$

Systematic dependence is introduced by making the intensity itself stochastic:

$$\Lambda \sim \text{Gamma}.$$

All exposures share the same intensity, which creates correlated default outcomes.

In CreditRisk+, the gamma distribution is used to model variation in a common default intensity. The gamma distribution is a continuous distribution defined for positive values and is suitable for describing skewed variation around a given mean value. The gamma distribution can be parameterized with a shape parameter α and a scale parameter β. For a gamma-distributed stochastic variable Λ it holds that

$$E[\Lambda] = \alpha\beta, \qquad \text{Var}(\Lambda) = \alpha\beta^2.$$

The shape parameter thus controls the relative dispersion, while the scale parameter determines the level scale.

A central measure in credit risk context is the so-called gamma variance, defined as the relative variance in the default intensity:

$$CV^2 = \frac{\text{Var}(\Lambda)}{E[\Lambda]^2}.$$

For a gamma distribution, this expression reduces to

$$CV^2 = \frac{\alpha\beta^2}{(\alpha\beta)^2} = \frac{1}{\alpha},$$

which shows that the gamma variance depends solely on the shape parameter and constitutes a dimensionless measure of the degree of systematic variation in the portfolio's common risk driver.

The gamma variance affects the extent to which default events cluster in adverse system states and thus has a decisive impact on the tail of the loss distribution and economic capital. The expected loss, however, is not affected, as it is determined by the mean value of the intensity.

2.3 Common Denominator

In large homogeneous portfolios, losses are driven entirely by the common risk driver:

- conditional PD in Vasicek,
- intensity in CreditRisk+.

3 Example: CreditRisk+ intuition

Consider 10 000 loans with PD 1%:

$$E[\Lambda] = 100.$$

System states correspond to different realizations of Λ:

- Favorable: $\Lambda \approx 60$
- Average: $\Lambda \approx 100$
- Adverse: $\Lambda \approx 160$

3.1 Parallel example: how systematic dependence arises in the Vasicek model

To illustrate the parallel with CreditRisk+, we now consider the same portfolio in the Vasicek model.

Assume again a portfolio with 10 000 identical loans with unconditional probability of default of 1%. The expected loss thus corresponds on average to 100 defaults per year.

In the Vasicek model, each borrower is assumed to default if

$$X_i = \sqrt{\rho}\, Z + \sqrt{1-\rho}\, \varepsilon_i < \Phi^{-1}(p),$$

where $Z \sim \mathcal{N}(0,1)$ is a common systematic factor and $\varepsilon_i \sim \mathcal{N}(0,1)$ is idiosyncratic risk.

System states and conditional PD Given a realized value of the systematic factor $Z = z$, the probability of default becomes the same for all loans and is given by

$$p(z) = \Phi\left(\frac{\Phi^{-1}(p) - \sqrt{\rho}\, z}{\sqrt{1-\rho}}\right).$$

The outcome in the portfolio can then be interpreted as a system state determined by z. To provide intuition, one can again consider three representative states:

- A favorable state with high z, where $p(z)$ is significantly lower than 1% and very few loans default.
- An average state around $z \approx 0$, where $p(z) \approx 1\%$ and the number of defaults is close to 100.
- An adverse state with low z, where $p(z)$ becomes strongly elevated and a large cluster of loans defaults.

Given $Z = z$, default outcomes in the portfolio are otherwise independent. Just as in CreditRisk+, it is therefore the variation in the common system state, not idiosyncratic randomness, that drives clustering of default.

4 Derivation of the relationship between ρ and gamma variance

The central relationship between the asset correlation ρ in the Vasicek model and the gamma volatility in CreditRisk+ can be derived by studying the variation in the common risk driver in the large portfolio limit.

4.1 Conditional PD as a random variable in the Vasicek model

In the Vasicek model, the conditional probability of default is given by

$$p(Z) = \Phi\left(\frac{\Phi^{-1}(p) - \sqrt{\rho}\, Z}{\sqrt{1-\rho}}\right),$$

where $Z \sim \mathcal{N}(0,1)$ is the systematic factor.

Since Z is random, $p(Z)$ is also a random variable. For low unconditional PD and moderate variations in Z, the function $p(Z)$ can be linearized around $Z = 0$:

$$p(Z) \approx p(0) + p'(0)\, Z.$$

The derivative is given by

$$p'(Z) = -\frac{\sqrt{\rho}}{\sqrt{1-\rho}}\,\phi\left(\frac{\Phi^{-1}(p) - \sqrt{\rho}\,Z}{\sqrt{1-\rho}}\right),$$

where ϕ is the density function of the standard normal distribution.

Evaluated at $Z = 0$, we obtain

$$p'(0) = -\frac{\sqrt{\rho}}{\sqrt{1-\rho}}\,\phi\left(\frac{\Phi^{-1}(p)}{\sqrt{1-\rho}}\right).$$

4.2 Approximation for low PD

For low values of p, the approximation holds[1]

$$\phi\left(\frac{\Phi^{-1}(p)}{\sqrt{1-\rho}}\right) \approx \frac{1}{\sqrt{1-\rho}}\,\phi\big(\Phi^{-1}(p)\big).$$

Inserted into the expression above gives

$$p'(0) \approx -\sqrt{\rho}\,\phi\big(\Phi^{-1}(p)\big).$$

The variance of the conditional PD can thus be approximated as

$$\mathrm{Var}[p(Z)] \approx (p'(0))^2\mathrm{Var}(Z) = \rho\,\phi\big(\Phi^{-1}(p)\big)^2.$$

Dividing this by the square of the unconditional PD gives the relative variance:

$$\frac{\mathrm{Var}[p(Z)]}{E[p(Z)]^2} \approx \rho\left(\frac{\phi(\Phi^{-1}(p))}{p}\right)^2.$$

For low PD, the ratio $\phi(\Phi^{-1}(p))/p$ grows at the same order as $(1-\rho)^{-1/2}$, which leads to the ASRF-established result

$$\frac{\mathrm{Var}[p(Z)]}{E[p(Z)]^2} \approx \frac{\rho}{1-\rho}.$$

4.3 Relative variance in CreditRisk+

In CreditRisk+, the default intensity Λ is assumed to be gamma distributed. For a gamma distribution, it holds exactly that

$$\frac{\mathrm{Var}(\Lambda)}{E[\Lambda]^2} = CV^2,$$

where CV^2 is the squared coefficient of variation, also called the gamma volatility.

[1] In Vasicek and ASRF-based portfolio models, terms of the type $\phi(c/\sqrt{1-\rho})$ appear, where $c = \Phi^{-1}(p)$ is the default threshold and ρ denotes the degree of systematic dependence. For low probabilities of default, $c \to -\infty$, which implies that asymptotic tail approximations for the normal distribution can be applied. In this regime, the approximation

$$\phi\left(\frac{c}{\sqrt{1-\rho}}\right) \approx \frac{1}{\sqrt{1-\rho}}\,\phi(c)$$

is often used. It is not exact but valid asymptotically for low default levels.

4.4 Identification in the large portfolio limit

In the large portfolio limit, the loss distribution is entirely driven by variation in the common risk driver. For the Vasicek model and CreditRisk+ to produce the same loss distribution, the relative variance of this risk driver must therefore be the same.

By identifying the stochastic PD in the Vasicek model with the intensity in CreditRisk+, we obtain

$$CV^2 \approx \frac{\rho}{1-\rho}.$$

Solving with respect to ρ yields the commonly used relationship

$$\rho \approx \frac{CV^2}{1+CV^2}.$$

This shows that the gamma volatility in CreditRisk+ is an (approximately) alternative parameterization of the same systematic credit risk as the asset correlation in the Vasicek model.

5 Calibrations

In this section, we consider two common calibration approaches linking the Vasicek model and CreditRisk+. The first approach matches the variance of the common risk driver, while the second matches a tail quantile such as the 99% Value-at-Risk.

5.1 Calibration via gamma variance and asset correlation

Assume that the asset correlation in the Vasicek model is $\rho = 10\%$. At a 99% confidence level, the outcome corresponds to a strongly negative realization of the systematic factor Z. The conditional PD then becomes significantly higher than 1%, which leads to a large cluster of defaults.

By exploiting the relationship derived earlier, we translate this into CreditRisk+ using

$$CV^2 \approx \frac{\rho}{1-\rho}.$$

In the numerical example,

$$\rho = 0.10.$$

Inserted into the expression above yields

$$CV^2 \approx \frac{0.10}{1-0.10} = \frac{0.10}{0.90} \approx 0.111.$$

By definition, the coefficient of variation satisfies

$$CV = \frac{\sqrt{\mathrm{Var}(\Lambda)}}{E[\Lambda]}.$$

Thus,

$$CV = \sqrt{CV^2} = \sqrt{0.111} \approx 0.333.$$

Economic interpretation This implies that the standard deviation of the common default intensity Λ is approximately 33% of its mean. At the 99% quantile, a high value of Λ is realized, leading to essentially the same number of defaults as in the Vasicek model. Numerical comparisons typically show that the 99% VaR differs only marginally between the two models.

5.2 Calibration of ρ to a given VaR level

In practical applications, it may be desirable not only to match the variance of the common risk driver, but to match a specific quantile of the loss distribution. This is particularly relevant if CreditRisk+ is used as the primary model and one wishes to determine a corresponding Vasicek correlation.

Problem formulation Assume that CreditRisk+ produces a 99% loss level (VaR) of

$$L_{0.99}^{CR+} = \ell_{99}.$$

We seek the correlation ρ in the Vasicek model such that

$$L_{0.99}^{Vasicek}(\rho) = \ell_{99}.$$

Large portfolio limit In a homogeneous portfolio with LGD = 1 and normalized exposures, the loss in the Vasicek model is

$$L(z) = p(z),$$

where z is the realization of the systematic factor.

The 99% loss corresponds to

$$L_{0.99}^{Vasicek} = p(z_{0.99}), \quad z_{0.99} = \Phi^{-1}(0.99).$$

Thus,

$$p(z_{0.99}) = \Phi\left(\frac{\Phi^{-1}(p) - \sqrt{\rho}\,\Phi^{-1}(0.99)}{\sqrt{1-\rho}}\right).$$

Solution for ρ We solve

$$\Phi\left(\frac{\Phi^{-1}(p) - \sqrt{\rho}\,\Phi^{-1}(0.99)}{\sqrt{1-\rho}}\right) = \ell_{99}.$$

Equivalently,

$$\frac{\Phi^{-1}(p) - \sqrt{\rho}\,\Phi^{-1}(0.99)}{\sqrt{1-\rho}} = \Phi^{-1}(\ell_{99}).$$

This yields

$$\sqrt{\rho} = \frac{\Phi^{-1}(p) - \Phi^{-1}(\ell_{99})\sqrt{1-\rho}}{\Phi^{-1}(0.99)}.$$

The equation is analytically solvable but is most conveniently handled numerically. Since the expression is monotonic in ρ, a unique solution exists.

Economic interpretation This calibration ensures that the Vasicek model reproduces the same extreme loss as CreditRisk+. It is particularly useful when CreditRisk+ is used for capital calculations but results need to be expressed in terms of asset correlation.

Despite differences in tail behavior, calibration to a specific quantile yields directly comparable risk measures across the two modeling frameworks.

6 Conclusion

The Vasicek model and CreditRisk+ are two representations of the same underlying systematic risk. With appropriate calibration, they produce nearly identical loss distributions in large portfolios, especially at high confidence levels such as 99%.